

Assignment 1

Log in to AWS

Create 2 Instances

1: Master

2: Slave

ON Master

1. sudo apt update (both for master and slave)
2. sudo apt install openjdk-17-jre -y (both for master and slave)
3. sudo nano install.sh

(Inside install.sh paste the installation commands for jenkins.

(You can find the command on jenkins official documentation. Copy the weekly release command.

```
sudo apt update
sudo apt install openjdk-11-jdk -y
sudo wget -O /usr/share/keyrings/jenkins-keyring.asc \
  https://pkg.jenkins.io/debian/jenkins.io-2023.key
echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] \
  https://pkg.jenkins.io/debian binary/ | sudo tee \
  /etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt-get update
sudo apt-get install jenkins -y
```

4. Check jenkins and Java on master

Java -version

Jenkins -version

5. Copy the public IP of the master, go to a new tab run it with port 8080
(Public IP: 8080)

6. Copy the path of jenkins where the password is given
(/var/lib/jenkins/secrets/initialAdminPassword)

7. Come back to your master
sudo cat /var/lib/jenkins/secrets/initialAdminPassword

8. Copy the password
9. Come back to jenkins dashboard > paste password> continue> install suggested plugins> write user name password> continue and finish.
10. Go on Manage Jenkins> System Configuration>Nodes>

Create Node

- Node Name: slave1> permanent agent
- Remote root directory:/home/ubuntu/jenkins/

Number of executors ?

1

Remote root directory ?

/home/ubuntu/jenkins/

Labels ?

- Launch method: via SSH

Dashboard > Manage Jenkins > Nodes >

Labels ?

Usage ?

Use this node as much as possible

Launch method ?

Launch agents via SSH

Launch agent by connecting it to the controller

Launch agents via SSH

172.31.0.156

Credentials ?

- none -

- Host: paste the private IP of slave1 (**test**)
- Credentials:> add> jenkins>

Dashboard > Manage Jenkins > Nodes >

Launch method ?

Launch agents via SSH

Host ?

172.31.0.156

Credentials ?

- none -

+ Add

Jenkins

credentials cannot be found

Host Key Verification Strategy ?

Known hosts file Verification Strategy

Kind: SSH username with private key

Jenkins Credentials Provider: Jenkins

Add Credentials

Domain

Global credentials (unrestricted)

Kind

- Username with password
- Username with password
- GitHub App
- SSH Username with private key**
- Secret file
- Secret text
- Certificate

Username ?

Username: ubuntu

Private key> enter directly: paste the PEM file

Username

ubuntu

☐ Treat username as secret ?

Private Key

☒ Enter directly

Key

No Stored Value

Passphrase

Click on ADD

☐ Treat username as secret ?

Private Key

☒ Enter directly

Key

Enter New Secret Below

```

06fvsGtbA+TEw3qMIpgjAhCFvmh0hq01p45j0DMNQpVGIEsGgqmDUYIqhsBP7f5W
WKH+vh/n1hYz4GG1Cqu/pb8t6muB0ykj1+o/48+FxZKUu3zHFMPy
-----END RSA PRIVATE KEY-----

```

Passphrase

Add Cancel

- Host key Verification: Non-verifying strategy

Host ?

172.31.0.156

Credentials ?

ubuntu

+ Add ▾

Host Key Verification Strategy ?

Non verifying Verification Strategy

Known hosts file Verification Strategy

Manually provided key Verification Strategy

Manually trusted key Verification Strategy

Non verifying Verification Strategy

Availability ?

Keep this agent online as much as possible

- Save > Refresh

Nodes

+ New Node Configure Monitors ↻

S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Linux (amd64)	In sync	4.85 GiB	0 B	4.85 GiB	0ms
	slave1	Linux (amd64)	In sync	5.32 GiB	0 B	5.32 GiB	90ms
Data obtained		0.32 sec	0.3 sec	0.26 sec	0.28 sec	0.26 sec	0.26 sec

Icon: S M L

11. Go to Slave1

```
ls
cd jenkins
ls
```

12. Go to Master

Create one folder to perform the tasks

```
mkdir git
```

```
cd git
git init
Touch masterfile
Git add .
Git commit -m "first commit for master"
Git branch develop
Git checkout develop
Touch developfile
Git add .
Git commit -m "commit for develop branch"
```

13. Go to GitHub

14. Create Repository

15. Copy the HTTPS Link of that repository

16. Go to the master

17. Git remote add origin link of repo

18. Git push -all

19. Now, Create a Job

New item> item Name: Job1 >Freestyle Project>Create

Dashboard > All >

Enter an item name

Job1

» Required field

Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by archiving artifacts and sending email notifications.

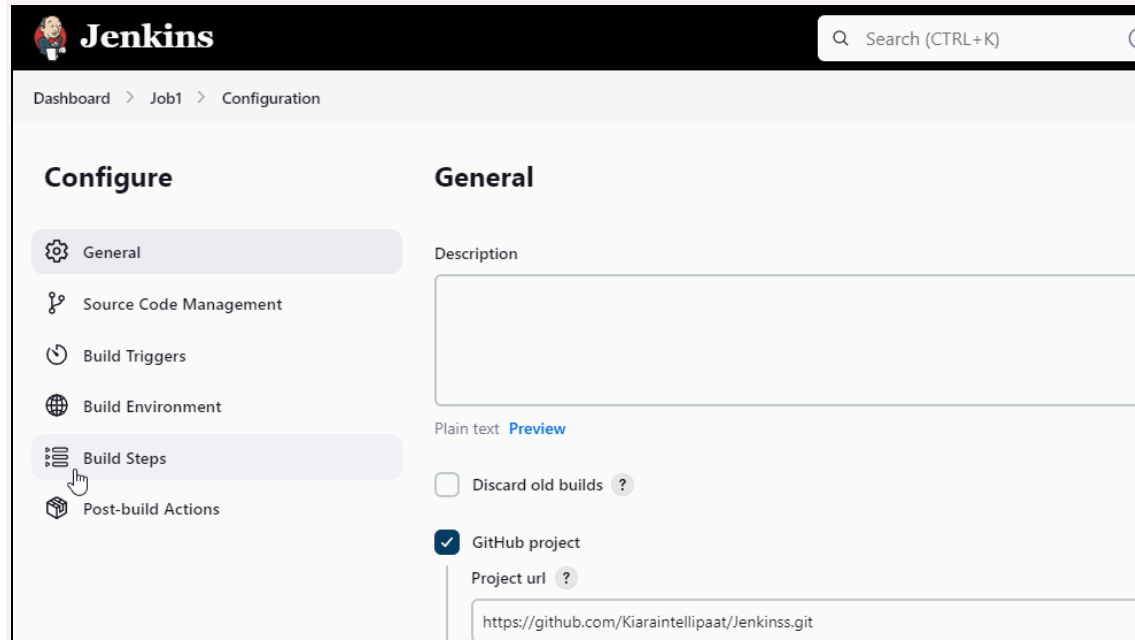
Pipeline
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formal) and/or organizing complex activities that do not easily fit in free-style job type.

Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, builds, etc.

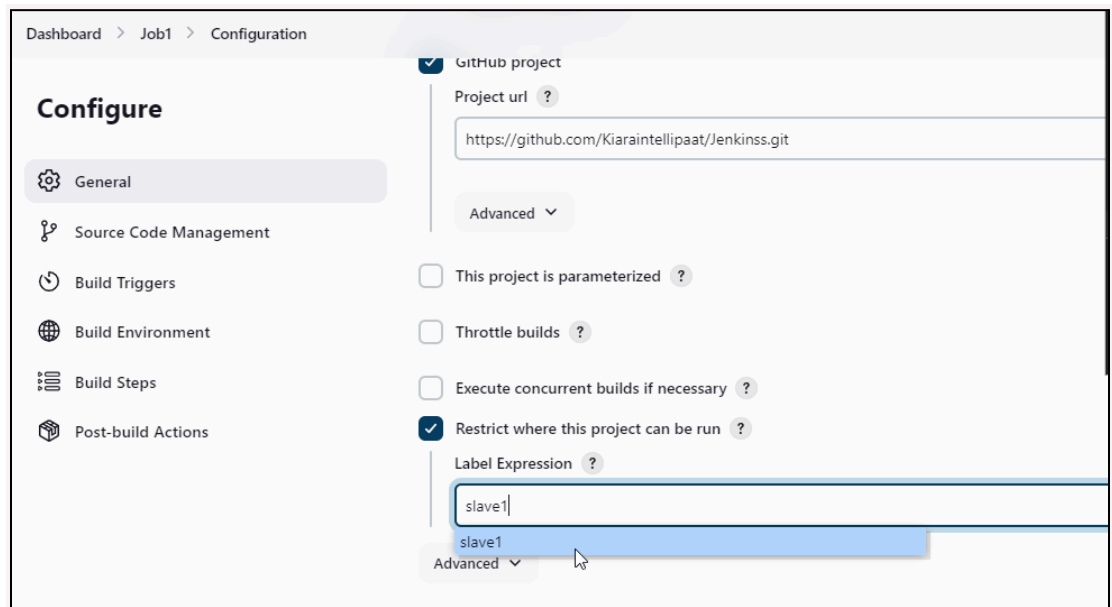
Folder
A container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a container for displaying items, you can have multiple things of the same name under the same in different folders.

OK

- Paste the GitHub repository link



- Click on restrict where project run: slave1 (Node name)



- ON Source Code Management
Github> paste url of the repo

Dashboard > Job1 > Configuration

Advanced ▾

Configure

- General
- Source Code Management
- Build Triggers
- Build Environment
- Build Steps
- Post-build Actions

Source Code Management

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/Kiaraintellipaati/Jenkins.git

! Please enter Git repository.

Credentials ?

- none -

- Specify Branch Name */develop

Dashboard > Job1 > Configuration

Configure

- General
- Source Code Management
- Build Triggers
- Build Environment
- Build Steps
- Post-build Actions

Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

*/develop

Add Branch

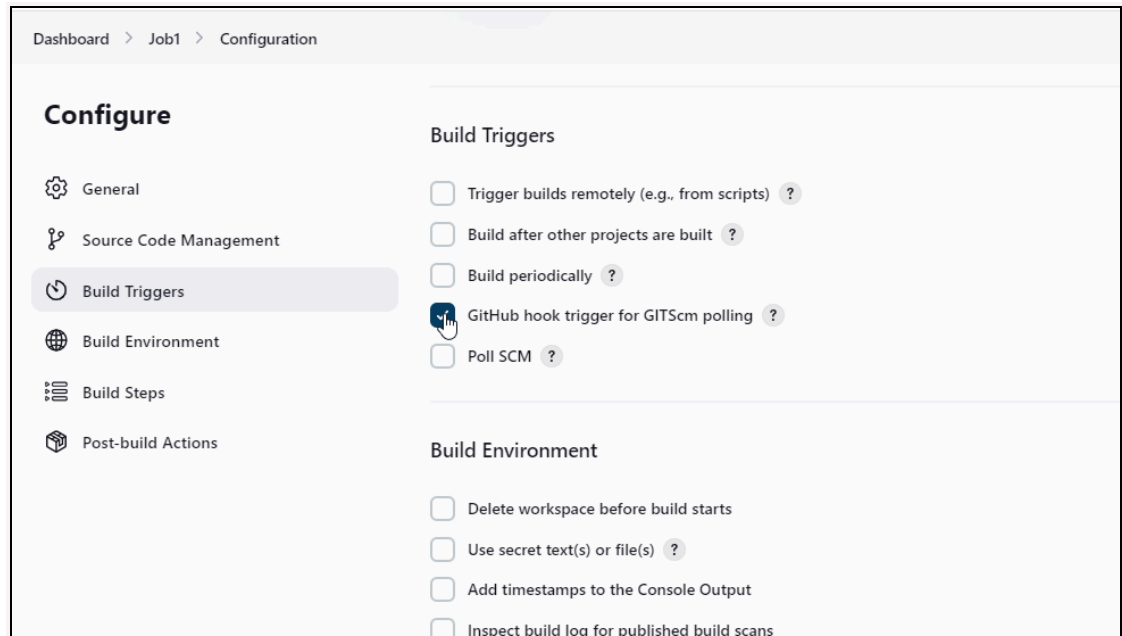
Repository browser ?

(Auto)

Additional Behaviours

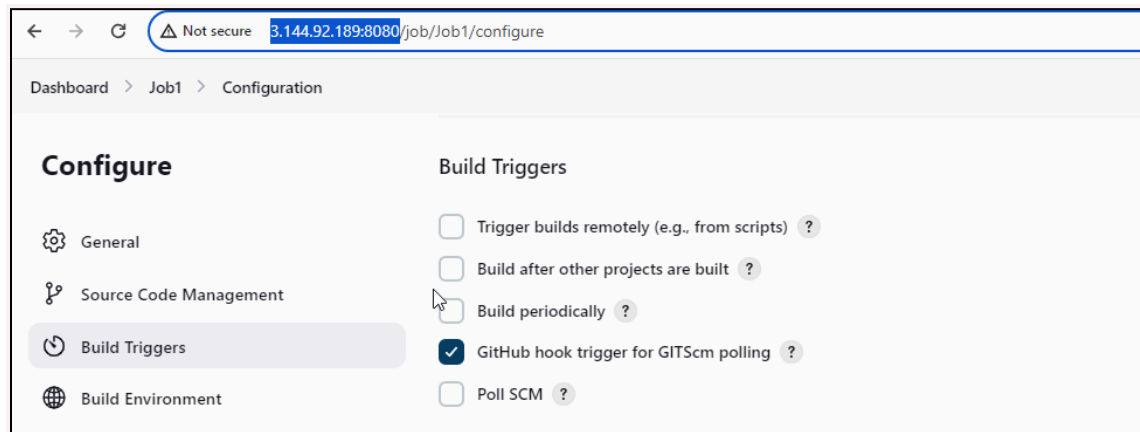
Add ▾

- Triggers> Github webhook triggers for GitSCM polling

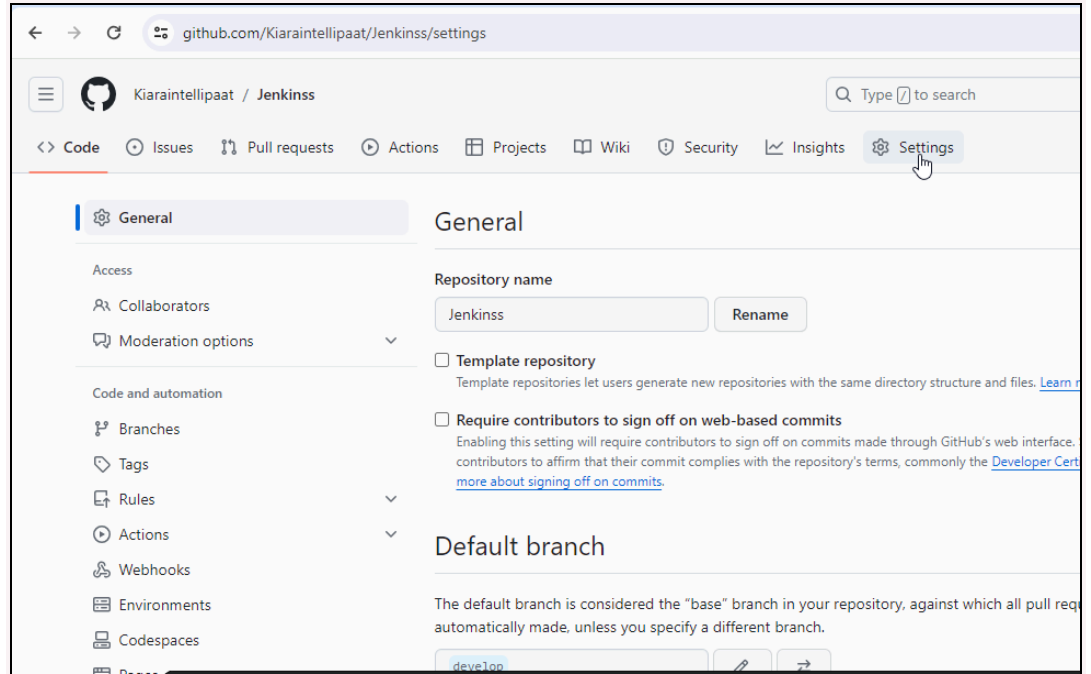


20. Create a Webhook:

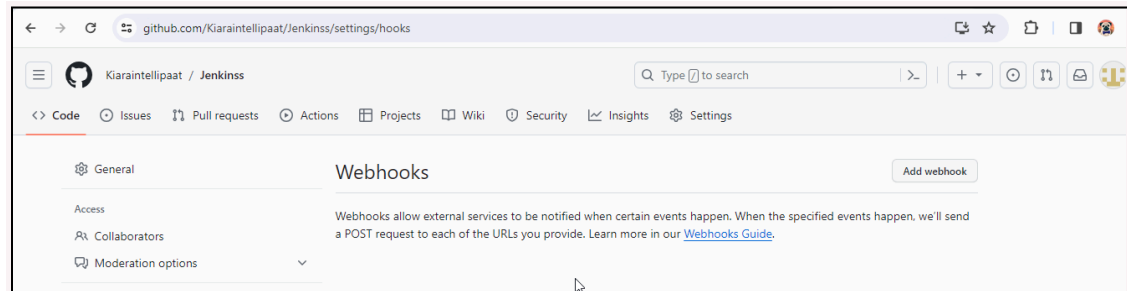
- Copy the Main Tab URL



- Go on GitHub of your repository
- Go on Settings



- Webhook>Add Webhook



- ON Payload URL
- -----URL-----/github-webhook/

- Click on Add Webhook

21. Now, Go Back on the Jenkins Dashboard > Save

22. Go on Build Now

23. Go on Slave

24. ls

25. cd workspace

26. ls

27. Job1

28. Now go back to Master

29. Touch newfile

30. Git add .

31. Git commit -m "committing newfile"

32. Git push origin master

33. Go to Jenkins Dashboard

Build 2 will be done automatically

Assignment 2

Create 2 more instances

Name: Prod

Name: Test

On both machine prod, test

1. sudo apt update
2. sudo apt install openjdk-11-jdk -y
3. Go to Manage Jenkins> System Configuration>Nodes>

Create Node: Prod

- Node Name: Prod> permanent agent
- Remote root directory:/home/ubuntu/jenkins/
- Launch method: via SSH
- Host: paste the private ip of slave2 (**prod**)
- Credentials:> add> jenkins>
- Kind: SSH username with private key
- Username: ubuntu
- Private key> enter directly: paste the PEM file>add

Host key Verification: Non-verifying strategy

Save > Refresh

4. Now, Create a Job

New item> item Name: Push to Prod>Freestyle Project>Create

- Paste the GitHub repository link
- Click on restrict where project run: Prod (Node name)
- On Source Code Management
- Github> paste url of the repo
- Specify Branch Name */develop
- Triggers> GitHub webhook triggers for GitSCM polling

Now build it

5. Go to Master

git checkout develop

Touch newfile

Git add .

Git commit -m "commit for newfile"

Git push origin develop

6. Go to Dashboard

2nd Build Done

7. Go To Prod

ls

```
cd jenkins
ls
cd workspace
cd Push to Prod
```

Similarly, for the next task:

Create Node

- Node Name: Test> permanent agent
- Remote root directory:/home/ubuntu/jenkins/
- Launch method: via SSH
- Host: paste the private IP of **Test**
- Credentials:> add> jenkins>
- Kind: SSH username with private key
- Username: ubuntu
- Private key> enter directly: paste the PEM file>add

Host key Verification: Non-verifying strategy

Save > Refresh

8. Now, Create a Job

New item> item Name: Push to Test>Freestyle Project>Create

- Paste the GitHub repository link
- Click on restrict where project run: Test (Node name)
- On Source Code Management
- Github> paste url of the repo
- Specify Branch Name */master
- Triggers> GitHub webhook triggers for GitSCM polling

Now build it

9. Go to Master

```
git checkout develop
```

```
Touch newfile
```

```
Git add .
```

```
Git commit -m "commit for newfile"
```

```
Git push origin develop
```

10. Go to Dashboard

2nd Build Done

11. Go To Test

```
ls
```

```
cd jenkins
```

```
ls
```

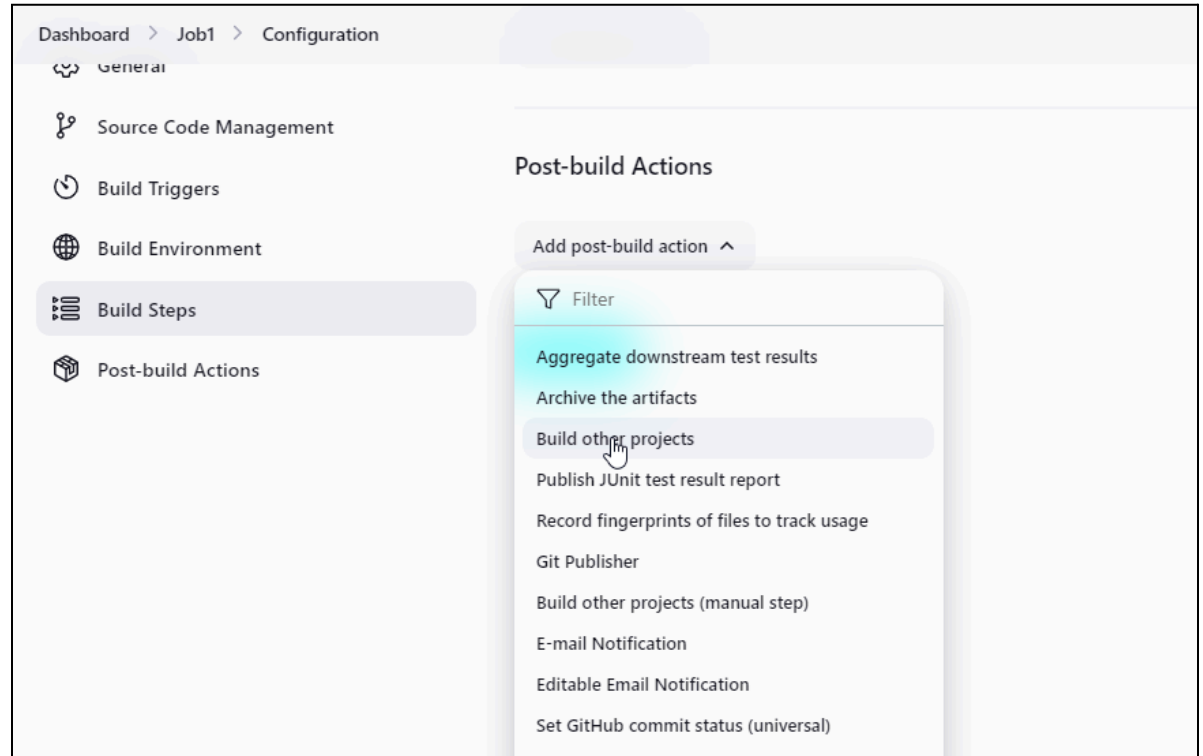
```
cd workspace
```

```
ls
```

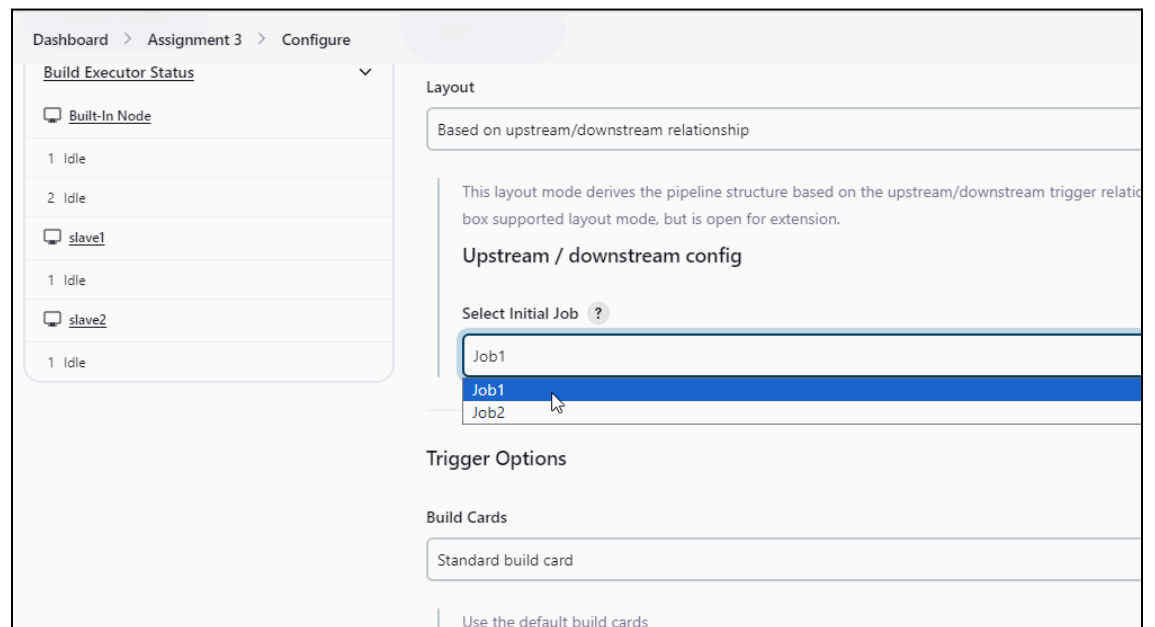
```
cd Push to Test
```

Assignment 3

1. Go to job Push to Test:
2. Change the branch name to develop
 - Add Post Build Action>Build other Project



- Project to Build: Push to Prod



3. Go to Manage Jenkins>Plugin Manager>Available Plugins

4. Search: Build Pipeline>Install
5. Create New View
 - Name: Assignment 3
 - Type: Build Pipeline View

Dashboard > New view

+ New Item

People

Build History

Project Relationship

Check File Fingerprint

Manage Jenkins

My Views

Build Queue

No builds in the queue.

Build Executor Status

Built-In Node

New view

Name

Assignment 3

Type

☒ Build Pipeline View

Shows the jobs in a build pipeline view. The complete pipeline of jobs that a version p are shown as a row in the view.

☐ List View

Shows items in a simple list format. You can choose which jobs are to be displayed in v

☐ My View

This view automatically displays all the jobs that the current user has an access to.

Create

- Apply
 - Save
6. Now go to the master machine, push the new file to the develop branch
 7. Your pipeline will be created and triggered automatically